

ASSIGNMENT 4 – CASE STUDY

PERERA D.S.

SLE320 -
RESOURCE
EFFICIENCY
AND WASTE
MANAGEMENT

INTRODUCTION

For this case study, I chose the café/restaurant sector because it produces multiple high-impact waste streams (food organics, packaging, glass, and general waste) and has clear opportunities for reduction and diversion. While the assessment requires a waste assessment and draft waste management plan for one organisation, I also conducted observations across two additional cafés/restaurants (with permission) to benchmark current practices, identify common gaps across the food industry, and strengthen the reliability of my conclusions. The final report is still centred on one primary site (China Bar), but the additional sites provide useful context to highlight recurring issues and support more practical, evidence-based recommendations.

I conducted on-site waste observations across the following three hospitality organisations:

- China Bar – restaurant
- Red cup café
- Subway - quick-service restaurant

Site details and audit setup (China Bar)

Organisation name: True Asian Kitchen

Address: 154 Victoria St, Richmond VIC 3121, Australia

Contact person + role: Kim Cheong Tan (Franchisee)

Date(s) of audit: 27th January 2026 , 28th January 2026

Audit type: Visual + volume estimates

Audit time window: 27th January 2026 (11 am – 11 pm), 28th January 2026 (6 pm – 11 pm)

Main services: Dine-in, Takeaway, Delivery, Bar

Peak periods: 12 pm – 3.30 pm, 6pm – 10.30 pm

Staff count (approx.): 12 (5/per shift)

Layout overview: key zones: kitchen, prep & dishwashing area, Bar & smoothie station, Front Of House, storeroom, outside bins

Site details and audit setup (café)

Organisation name: Red cup café

Address: 1124 Whitehorse Rd, Box Hill VIC 3128, Australia

Contact person + role: Rob Chen (owner)

Date(s) of audit: 28th January 2026

Audit type: Visual + volume estimates

Audit time window: 9.30 am – 1.30 pm

Main services: Dine-in, Takeaway, Delivery, Bar

Peak periods: 11 am – 3 pm

Staff count (approx.): 5/per shift

Layout overview: key zones: kitchen & dishwashing area, Bar & Barista station, Front Of House, storeroom, outside bins

Site details and audit setup (Subway)

Organisation name: Subway

Address: Shop 1 York Road, Mount Evelyn VI

Contact person + role: Gimhani rashenka Siriwardana (Shift Manager)

Date(s) of audit: 29th January 2026

Audit type: Visual + volume estimates

Audit time window: 12 pm – 2.30 pm

Main services: Dine-in, Takeaway, Delivery

Peak periods: 11 am – 3 pm

Staff count (approx.): 3/per shift

Layout overview: key zones: kitchen & dishwashing area, Front Of House, storeroom, outside bins

The food industry is one of the most visible and persistent generators of waste in modern cities because it operates through high-volume material flows that are tightly linked to time, hygiene, and customer expectations. Across cafés, restaurants, and fast-food outlets, waste is produced at every stage of service: food preparation generates organic offcuts; spoilage and overproduction create avoidable food loss; and front-of-house service produces large quantities of single-use packaging, cups, napkins, and mixed recyclables. In addition, deliveries introduce significant cardboard and plastic wrap, while cleaning and maintenance activities add further waste streams such as used oils, contaminated materials, and occasionally hazardous items. As a result, hospitality businesses often manage multiple waste streams simultaneously, yet the pace of service and limited back-of-house space can make correct separation difficult, increasing contamination and sending recyclable or compostable materials to landfill.

This matters because waste in the food industry is not simply a disposal problem—it is an efficiency, cost, and environmental issue. When valuable materials are discarded or contaminated, businesses pay more for collection and landfill charges while losing opportunities for diversion through recycling, organics processing, or reuse systems. Environmentally, landfilled organic waste contributes to greenhouse gas emissions and resource depletion, while mismanaged packaging waste increases pollution and undermines recycling performance. Addressing food-industry waste therefore requires moving beyond “bin-focused” solutions to a more systems-based approach that prioritises waste prevention, improved separation, staff training, and practical infrastructure design. A structured waste assessment and waste management plan provides an evidence-based pathway to identify where waste is generated, how it is currently handled, and what targeted changes can reduce landfill, improve diversion outcomes, and support broader sustainability goals.

METHODOLOGY AND DISCUSSION

Data Collection

A visual waste audit was conducted to understand how waste is generated, separated, and managed across the site. The audit began with a walkthrough of all waste-generating areas (kitchen, prep & dishwashing area, bar and smoothie station, front-of-house service area) and the external waste storage area.

During the walkthrough, each waste stream present at the site was identified (general waste, commingled recycling, cardboard, glass, organics/food waste, cooking oil). For each bin encountered, the bin type and size were recorded (or calculated where size labels were missing), along with its location and who primarily uses it (FOH staff vs Kitchen staff)



FOH Bins



External waste storage area

Bin size

large – general waste bin: 24.4 L

Small – Empty Bottle Collection Bin: 15.9 L



Bin size: 76 L

Prep & Dishwashing area / Kitchen Bins



Toilet Bin (Bin size: 18.2 L)

The contents of each bin were visually inspected without manual sorting, and key contamination indicators were noted.

Bin fullness was estimated using consistent categories ($\frac{1}{4}$, $\frac{1}{2}$, $\frac{3}{4}$, full) and cross-checked by observing how quickly bins filled during service periods and asking staff (where appropriate) about emptying frequency and peak-time overflow.

FOH Bin (General waste Bin)

Date	27/01/2026
Time	11.08 am (Just after opening of the restaurant)
Area/station	Front of House – next to the smoothie station
Waste stream	General waste
Bin type	Bucket style general waste bin (It appears to be a portable plastic utility pail rather than a standard wheelie bin)
Bin size (L)	• Height $h = 37$ cm • Diameter = 29 cm $\Rightarrow$ radius $r = 14.5$ cm
Volume calculation	$V = \pi r^2 h = \pi (14.5^2)(37) = \pi (210.25)(37) = 7779.25\pi \text{ cm}^3 \approx 24,439 \text{ cm}^3$ Since $1 \text{ cm}^3 = 1 \text{ mL}$ ✓ Volume $\approx 24,439 \text{ mL}$ ✓ $\approx 24.4 \text{ L}$
Fullness ($\frac{1}{4}$ / $\frac{1}{2}$ / $\frac{3}{4}$ /full)	$\frac{1}{4}$ full
Liners? (Yes/No)	Yes
Lid closed? (Yes/No)	No
Visible contamination	No obvious “wrong-stream” items are visible yet, so contamination appears low at this time point.
Main contaminants seen	At the start of service (restaurant opened at 11:00 am), the bin is almost empty, which makes sense because waste generation has only just begun. The visible items are mainly blue disposable gloves (PPE) and paper/tissue/paper towel.

Notes (odour/leaks/sharps/pests)	This setup has a high risk of future contamination/missed diversion because: • The bin is unlabelled. • It's an open bin in a convenient work area—so staff may drop recyclables or food scraps into it during busy periods.
Picture	

Date	27/01/2026
Time	8 pm
Area/station	Front of House – next to the smoothie station
Waste stream	General waste
Fullness (1/4/1/2/3/4/full)	3/4 full
Liners? (Yes/No)	Yes
Lid closed? (Yes/No)	No
Visible contamination	The bin contains items that could potentially be diverted from general waste depending on the site's available streams and accepted materials—most notably a plastic beverage cup/lid and straw. The rest appears dominated by paper towels/napkins, which are typically not recyclable and would only be divertible if an organics/FOGO stream is available and permitted by the contractor.
Main contaminants seen	Potential recyclables in general waste: a rigid plastic drink cup/lid (and straw). Food-soiled paper: multiple used napkins/paper towels (not suitable for recycling; may belong in organics if accepted). Residual liquid risk: the cup appears to contain or have contained liquid, which increases wet/soiled waste in the general stream.
Notes (odour/leaks/sharps/pests)	Wet/soiled waste risk: Residual beverage/liquid can create minor leachate, make liners messy, and increase odour over time even indoors. Pests can still occur indoors (flies/ants/cockroaches) because they can enter via doors opening, deliveries, small gaps, and drains—food/drink residues are a strong attractant.

Picture	
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FOH Bin (Empty bottle collection bin)

Date	27/01/2026
Time	12.33 pm, 7.33 pm
Area/station	Front of House – next to the General waste bin
Waste stream	Beverage container collection (internal recycling stream): used to collect empty drink containers (e.g., glass bottles / plastic bottles / aluminium cans) before transferring to the main recycling system/collection point.
Bin type	Portable plastic utility pail/ plastic bucket
Bin size (L) Volume calculation	• Height $h = 21$ cm • Diameter $d = 31$ cm $\rightarrow$ radius $r = 15.5$ cm $V = \pi r^2 h = \pi (15.5^2)(21)$ $15.5^2 = 240.25$ $240.25 \times 21 = 5045.25$ $V = 5045.25\pi \approx 15,850.12 \text{ cm}^3$ Since $1 \text{ cm}^3 = 1 \text{ mL}$: Volume = 15,850 mL = 15.85 L
Fullness ($\frac{1}{4}$ / $\frac{1}{2}$ / $\frac{3}{4}$ /full)	12:33 pm: empty. 7:33 pm: $< \frac{1}{4}$ full (only a few items visible).
Liners? (Yes/No)	No
Lid closed? (Yes/No)	No
Visible contamination	Low contamination risk: all contents were beverage containers
Main contaminants seen	Beverage carton; potential residual liquid in containers (can cause minor leakage even if no food waste is present).
Notes (odour/leaks/sharps/pests)	No obvious food waste in the bin; low odour expected, but spill/leak risk if containers aren't emptied. Even indoors, pests can still be attracted if sugary residues are present—especially if the lid is left open for long periods. No obvious sharps; however, crushed cans can create sharp edges.

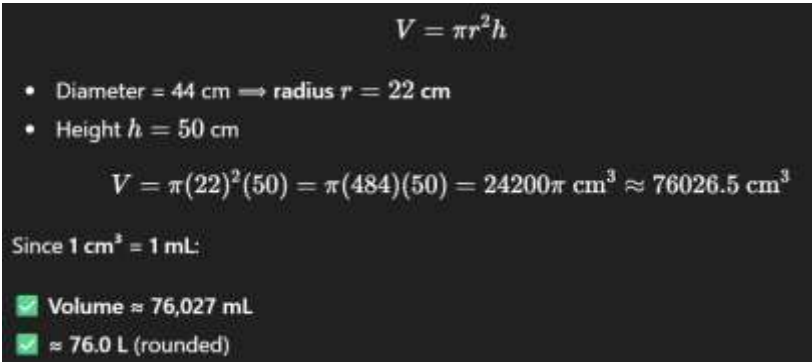
Picture	 12.33 pm 7.33 pm
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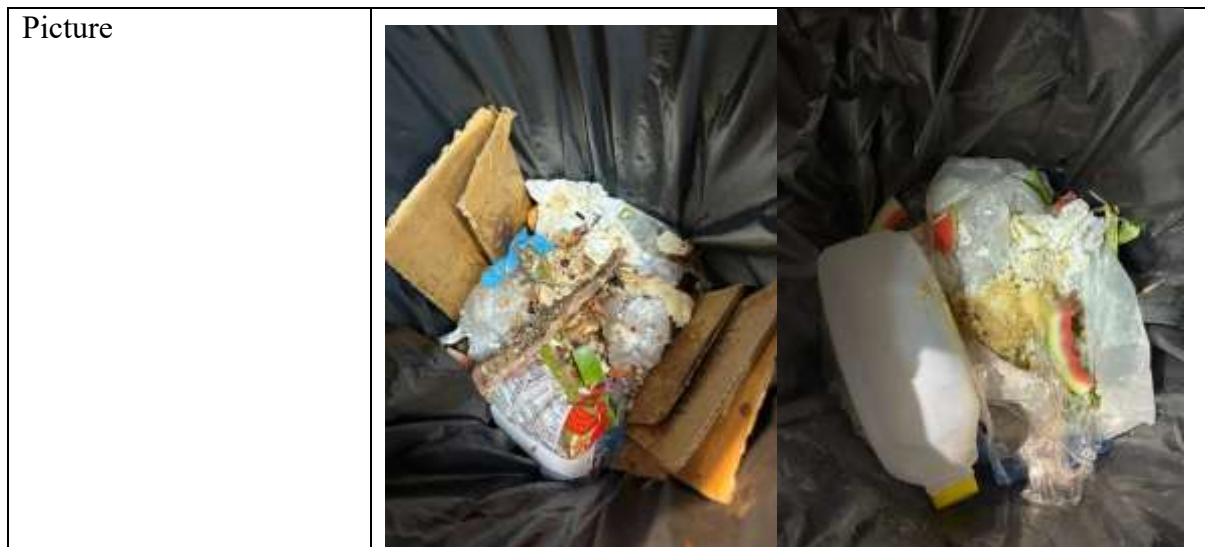
Date	28/01/2026
Time	10.51 pm
Fullness (1/4/1/2/3/4/full)	Full / overflowing
Liners? (Yes/No)	No
Lid closed? (Yes/No)	No
Visible contamination	Mostly “correct” stream (glass bottles + aluminium cans), but not well contained (overflow/stockpiled around bin) and likely residual liquid in containers.
Main contaminants seen	No obvious non-target materials observed (contents appear consistent with intended stream: glass + aluminium + plastic bottles). Bottle lids/caps are being placed in general waste, even when the bottles are being collected separately.

	
Notes (odour/leaks/sharps/pests)	Overflow/stockpiling indicates insufficient capacity or emptying frequency on busy nights. Breakage/sharps risk (glass bottles on floor). ‘Slip’ hazard from bottles stored around the bin and possible leaks/sticky spills from unemptied cans/bottles. Even indoors, pests can still occur because drink residue + warm indoor areas + nearby drains can attract them if waste is left overnight. Most lids are plastic or aluminium (and the metal rings are recyclable), so putting them in general waste reduces diversion and increases landfill-bound material.
Picture	

Kitchen Bins

Date	27/01/2026
Time	1.54 pm
Area/station	Kitchen area
Waste stream	Kitchen food waste / organics
Bin type	Open-top round rigid plastic bin / utility bin
Bin size (L) Volume calculation	$V = \pi r^2 h$ - Diameter = 44 cm $\Rightarrow$ radius $r = 22$ cm - Height $h = 50$ cm $V = \pi(22)^2(50) = \pi(484)(50) = 24200\pi \text{ cm}^3 \approx 76026.5 \text{ cm}^3$ Since $1 \text{ cm}^3 = 1 \text{ mL}$: ✅ Volume $\approx 76,027 \text{ mL}$ ✅ $\approx 76.0 \text{ L}$ (rounded)
Fullness ($\frac{1}{4}$ / $\frac{1}{2}$ / $\frac{3}{4}$ /full)	$\frac{1}{4}$ full
Liners? (Yes/No)	Yes (black bin liner)
Lid closed? (Yes/No)	No
Visible contamination	Yes
Main contaminants seen	The contents are mostly organics, however soft plastic/film packaging is visible, which contaminates an organics stream and can reduce recovery quality. And minor non-food items such as tissues are visible.
Notes (odour/leaks/sharps/pests)	Food waste is wet/food-soiled, so there is leachate (drip) + odour risk if the bin is left open or not emptied regularly. Even indoors, pests can still be attracted if food residues are present—especially with an open lid. No obvious sharps visible in the photo.
Picture	

Date	27/01/2026
Time	8.30 pm – 8.33 pm
Area/station	Kitchen area
Bin type	Open-top round rigid plastic bin / utility bin
Bin size (L) Volume calculation	 $V = \pi r^2 h$ - Diameter = 44 cm $\Rightarrow$ radius $r = 22$ cm - Height $h = 50$ cm $V = \pi(22)^2(50) = \pi(484)(50) = 24200\pi \text{ cm}^3 \approx 76026.5 \text{ cm}^3$ Since $1 \text{ cm}^3 = 1 \text{ mL}$: ☒ Volume $\approx 76,027 \text{ mL}$ ☒ $\approx 76.0 \text{ L}$ (rounded)
Waste stream	Kitchen – General waste (residual waste) (mixed kitchen refuse not separated into recycling/organics).
Fullness ($\frac{1}{4}$ / $\frac{1}{2}$ / $\frac{3}{4}$ /full)	$\frac{1}{2}$ full
Liners? (Yes/No)	Yes
Lid closed? (Yes/No)	No
Visible contamination	Moderate–high diversion opportunity (i.e., “contamination” of general waste by recoverable materials): visible cardboard, rigid plastic container (large bottle/jerrycan-type), soft plastics/film, and significant food scraps/organics (e.g., watermelon rind and mixed leftovers). These materials indicate separation is not happening consistently at the point of disposal.
Main contaminants seen	Food/organics: watermelon rind + mixed food residues/scraps. Recyclables: cardboard pieces. Plastics: large rigid plastic container + plastic bags/film/packaging. (These are “contaminants” only in the sense that they could be diverted from landfill if recycling/organics systems are available and followed.)
Notes (odour/leaks/sharps/pests)	Wet/food-soiled waste is present, so odour + leachate (leak) risk is moderate—especially if liquids remain in containers or food waste sits for long periods. Even indoors, pests can still be attracted, particularly around warm areas, drains, and food residue. No obvious sharps are visible, but bones/hard food waste and rigid packaging edges can create minor handling risk.



Prep & Dishwashing Area Bins

Date	27/01/2026
Time	7.56 pm
Area/station	Dishwashing area bin (plate-scraping)
Bin type	Open-top cylindrical plastic kitchen/utility bin
Bin size (L)	
Volume calculation	
Waste stream	Food residues / plate-scraping waste (pre-rinse solids removed before dishes enter dishwasher). The site has no organics service, this stream is effectively being managed as general waste even though it is food-dominant.
Fullness ($\frac{1}{4}$ / $\frac{1}{2}$ / $\frac{3}{4}$ /full)	$\frac{3}{4}$ full
Liners? (Yes/No)	Yes
Lid closed? (Yes/No)	No
Visible contamination	Low contamination risk Contents appear mostly food residues and food-soiled paper towels/napkins, which match the intended purpose of a plate-scraping bin.
Main contaminants seen	Food-soiled paper towels/napkins mixed with food scraps. (These may be acceptable in some organics systems, but rules vary. If this bin is meant strictly for food only, paper adds volume and moisture and can reduce efficiency.)
Notes (odour/leaks/sharps/pests)	Waste here is wet/food-soiled, so there is a higher odour + microbial growth potential, especially in a warm dishwashing area. Even indoors, pests can still occur because kitchens have doors opening, deliveries, floor drains, gaps, and residual food moisture. Leak risk is also higher due to liquids and sauces. No obvious sharps seen, but general handling hygiene is important.

Picture



Prep Area Bin

Date	27/01/2026
Time	3.25 pm
Area/station	Food preparation Area
Waste stream	Food residues
Bin type	Open-top cylindrical plastic kitchen/utility bin
Bin size (L)	
Volume calculation	$V = \pi r^2 h$ - Diameter = 44 cm $\Rightarrow$ radius $r = 22$ cm - Height $h = 50$ cm $V = \pi(22)^2(50) = \pi(484)(50) = 24200\pi \text{ cm}^3 \approx 76026.5 \text{ cm}^3$ Since $1 \text{ cm}^3 = 1 \text{ mL}$: ✓ Volume $\approx 76,027 \text{ mL}$ ✓ $\approx 76.0 \text{ L}$ (rounded)
Fullness ($\frac{1}{4}/\frac{1}{2}/\frac{3}{4}/\text{full}$)	$\sim \frac{1}{2}$ full (moderate load at time of photo).
Liners? (Yes/No)	Yes
Lid closed? (Yes/No)	No
Visible contamination	Yes: mixed materials visible in one bin (food prep residues + packaging materials), indicating stream-mixing risk.
Main contaminants seen	Food-soiled aluminium foil, vegetable offcuts (e.g., spring onion pieces), and possible packaging/beverage container mixed in the same stream.
Notes (odour/leaks/sharps/pests)	Odour: likely moderate (food/oil residue). Leak risk: moderate due to moist/oily waste and soiled foil. Sharps: no obvious sharps seen in this photo. Pests: possible if left open or not emptied regularly.
Picture	

Restroom bins: two units, located in the men's and women's toilets

Date	27/01/2026
Time	7.42 pm
Area/station	Inside the restrooms
Waste stream	General waste (restroom residual waste)
Bin type	Small indoor rectangular plastic utility bin
Bin size (L)	$V = L \times W \times H$ $V = 26 \times 20 \times 35 = 520 \times 35 = 18,200 \text{ cm}^3$ Since $1 \text{ cm}^3 = 1 \text{ mL}$: ✓ Volume = 18,200 mL ✓ = 18.2 L
Volume calculation	
Fullness ($\frac{1}{4}$ / $\frac{1}{2}$ / $\frac{3}{4}$ /full)	$\frac{1}{4}$ (low fill)
Liners? (Yes/No)	Yes
Lid closed? (Yes/No)	No
Visible contamination	Low — no obvious cross-stream contamination visible
Main contaminants seen	Mostly used tissues/paper towels; possible minor moisture/soiling on tissues
Notes (odour/leaks/sharps/pests)	No obvious sharps or leakage visible in the photo. No pests observed. As this is an open indoor restroom bin, regular emptying and liner replacement are recommended to control odour and maintain hygiene.
Picture	

Bin access time:

FOH staff take approximately 11 seconds to reach FOH bins and 15 seconds to reach dishwashing-area bins, while kitchen staff take around 4 seconds to reach kitchen bins. This suggests bin placement is most efficient in the kitchen and comparatively less efficient for FOH/dishwashing tasks, which may reduce correct disposal compliance during busy periods.

Staff interview finding :

“When asked which menu item generates the highest amount of waste, the chef identified duck. This is mainly due to its low edible yield compared with purchase weight. Duck preparation produces substantial trim waste (skin, fat, bone, and connective tissue), and portioning can leave additional off cuts. In service, duck dishes may also produce higher plate waste if portions are large or if the dish is rich, increasing leftovers.”

Unhealthy customer practice observed:

“Some customers left duck bones on dining tables instead of placing them in plates”

Why this matters:

- Increases hygiene risk on shared surfaces
- Can attract pests if not cleared quickly
- Adds cross-contamination risk during table resetting
- Increases FOH cleaning workload and turnaround time

Unhealthy staff practice observed:

“Restroom tissues are removed from restroom bins, but the bin liner is not replaced. The collected tissues are then disposed of in the bin near the dishwasher area.”

Why this is a problem:

- Increases cross-contamination risk between restroom waste and food-service areas
- Raises odour and hygiene concerns in the dishwashing zone
- Can transfer microbes via hands, bin surfaces, and shared equipment
- Does not follow good waste-segregation and sanitation practice

Other practices:

Reusable carry bags (hanging near service area)

This is a good sustainability practice. The bags are clearly labeled as reusable and encourage customers to reuse instead of taking a new bag each visit. That can reduce single-use bag waste over time. The only limitation is that they are still plastic-based, so the benefit depends on customers actually reusing them multiple times.



Disposable takeaway container stock (clear tubs/lids/cups)

This is a necessary but high-waste practice. Keeping takeaway containers ready supports operations, but these items are mostly single-use and can generate large plastic waste volumes, especially in busy periods.

Charging \$0.50 per takeaway container helps reduce unnecessary single-use packaging because it makes customers think before requesting one. In waste terms, this can lower the number of plastic tubs/lids entering general waste, especially for dine-in customers who might otherwise take a “just in case” container. So this is a positive waste-minimisation measure.

To make it stronger, the restaurant could pair the fee with a clear message such as: “Bring your own clean container” or “No cutlery unless requested,” and track how many containers are sold each week. If container sales drop over time, it’s good evidence that the charge is reducing single-use waste rather than just adding cost.



Cutlery/utensil setup (small wooden spoons/chopsticks in holder)

This is a mixed practice. Wooden or plant-based utensils are generally better than fossil-plastic utensils, but if they are single-use, they still create avoidable waste. The wrapping around each item improves hygiene, but adds extra packaging waste.

The venue provides single-use utensils only when customers request them, which helps prevent unnecessary distribution and reduces avoidable waste. For dine-in service, staff use reusable metal cutlery that is washed, sanitised, and reused, which aligns well with waste-minimisation principles and lowers single-use material consumption. Overall, this setup balances hygiene and customer choice while keeping packaging waste relatively low.



Individually wrapped straws (boxed stock)

This is also mixed. Individual wrapping supports hygiene and customer confidence, but creates additional waste per item. These are paper/plant-fiber straws, so that is better than conventional plastic straws. Best improvement: keep them unwrapped where food-safety rules allow, or provide wrapped items only upon request rather than automatic distribution.



“No outside food or drink” sign

From a waste perspective, this is neutral to slightly positive operationally, but not a major sustainability control by itself. It may reduce mixed external packaging and contamination inside bins, which helps cleaner sorting. However, it does not directly reduce the restaurant’s own packaging footprint. The stronger waste impact will come from bin segregation, packaging choices, and staff serving habits.



FOH staff currently separate cardboard from other packaging waste, which shows good source-separation behaviour and awareness of recycling. However, because there is no dedicated cardboard bin at the point of generation, the material is temporarily stored at the back of the prep area. This creates an informal storage system that can lead to overflow, contamination, and handling inefficiencies. If cardboard becomes wet or food-stained, recyclability drops.



At China Bar, staff empty all bins daily. Liners are replaced with new ones for every bin except those in the restrooms; restroom bins are emptied daily, but their liners are not changed.

Oil residue management (used cooking oil/fat residues):

This is a good practice because oil residues are being collected separately instead of being poured into sinks or mixed with general waste. Separating oil helps prevent drain blockages, reduces grease-related odour/pest issues, and supports better waste handling.



- Terms & Conditions:**
1. Check all goods on delivery. Report any damage or discrepancy within 5 working days.
 2. Returns must be authorised in writing and made within 10 working days. Unauthorised or damaged returns will not be credited.
 3. Custom or special-order goods are non-returnable.
 4. All goods remain the property of ACOPACK PTY LTD until paid in full.
 5. Liability is limited to replacement or refund at our discretion for claims made within 10 working days.

The business owner confirmed that the restaurant's waste contractor is Waste Management Pty Ltd (trading as Urban Waste). This is consistent with the account statement/invoice records for CHINA BAR, which list Urban Waste as the service provider and provide Christina's contact details.

To verify operational details, I phoned Christina using the invoice contact; she referred me to Shane Maletzki (Sales Manager, Urban Waste | shane@urbanwaste.com.au). I then emailed Shane with questions about the waste collection procedure, and his responses were used to confirm how collection is currently managed on site.



STATEMENT

CHINA BAR
C/- WT 99 PTY LTD
154 VICTORIA STREET
RICHMOND VIC 3121
AUSTRALIA

As At
21 Jan 2026

ABN
60 350 832 630

Waste Management Pty
Ltd T/As Urban Waste
300 City Rd
SOUTHBANK VIC 3006
AUSTRALIA
Email:
christina@urbanwaste.com.au
Phone: 0427869461

Date	Activity	Reference	Due Date	Invoice Amount	Payments	Balance AUD
31 Dec 2025	Invoice # 30374	544	14 Jan 2026	198.99	0.00	198.99

BALANCE DUE AUD 198.99

Terms: 14 days from invoice date.

NAB - NATIONAL AUSTRALIA BANK LIMITED
ACCOUNT: WASTE MANAGEMENT PIL ATF WASTE MANAGEMENT TRUST
BSB: 083-166
ACCOUNT NO: 937274879

PAYMENT ADVICE

To: Waste Management Pty Ltd T/As Urban Waste
300 City Rd
SOUTHBANK VIC 3006
AUSTRALIA
Email: christina@urbanwaste.com.au
Phone: 0427869461

Customer	CHINA BAR	
Overdue	Current	Total AUD Due
198.99	0.00	198.99
Amount Enclosed		
	Enter the amount you are paying above	



(Cardboard Bin)



(General waste bin)

External waste service confirmation (Urban Waste)

Waste collection services for True Asian Kitchen/ China Bar are provided by Urban Waste (private contractor). Service details were confirmed through direct communication with the provider team: an initial phone call with Christina (contact listed on invoice), followed by referral to Shane Maletzki (Sales Manager), who confirmed operational details by email.

The account currently includes two waste streams: general waste and cardboard. Bin infrastructure on the account is 1 × 660 L general waste bin and 1 × 660 L cardboard bin.

Collections are run on a fixed schedule (not on-call). The current frequency is:

- General waste: weekly (Tuesday)
- Cardboard: fortnightly (Tuesday)

The provider advised that there have been no recent missed services. However, they also confirmed that pickup weight data is not available (no onboard weighing/stream weight reporting), which limits quantitative performance tracking for this site.

After collection, general waste is sent to Dynon Road landfill, while cardboard is sent to Visy (Springvale) for processing.

Any post-collection sorting is handled at the receiving facility. Urban Waste also advised that they can provide cardboard-only pickups and baling options (compressing loose cardboard into tight, tied bundles (“bales”) using a baler machine because takes up much less space, fewer pickups needed, can reduce collection costs for high-cardboard site) if cardboard volumes increase.

LIMITATIONS AND ACTIONS

A staff member separately collects eligible drink containers and redeems them through CDS Vic for the 10-cent refund per container, diverting these items from general waste.”

In Victoria, the 10-cent system is the Container Deposit Scheme (CDS Vic). People return eligible drink containers at CDS return points (e.g., depots, reverse vending machines, and over-the-counter points) and receive a refund.

Not every container is eligible. In general, eligible containers are beverage containers in the scheme range (commonly 150 mL to 3 L) with the correct CDS refund marking. Some containers are excluded, including plain milk containers, flavoured milk containers 1 L and above, and wine/spirit bottles. (Victoria’s Container Deposit Scheme | Find Eligible Containers, n.d.)



Since wine bottles are clearly visible here, the best action is to run a 2-stream bottle setup:

- Stream 1 (CDS refund): eligible cans/bottles with the 10c mark (and barcode), kept empty and not crushed.
- Stream 2 (non-CDS glass): wine/spirit bottles, collected separately for recycling (not CDS).

Because the current collection setup is general waste + cardboard, the wine bottles are collected through the urban waste cardboard stream and sent to Visy Recycling in Springvale.

As seen in the photo, a milk carton is being placed in the general waste stream because it is not eligible for CDS refund. During my investigation at Red Cup Café, I observed that milk cartons are stored separately, and a milk supplier collects them for reuse. China Bar could adopt the same practice by separating milk cartons from general waste and arranging separate collection, so these containers are diverted from landfill.



(China Bar)



(Red Cup Café)

A major issue identified at China Bar is the absence of waste-bin signage across the site. Without clear labels (what goes in / what stays out), staff are forced to guess disposal decisions during busy periods, which increases contamination between streams (especially general waste vs recyclables/cardboard). This reduces recovery opportunities, increases landfill disposal, and can create hygiene risks when food-soiled items are placed in incorrect bins and left for longer periods.

A practical solution is to implement a standardised signage system on every bin, similar to what was observed at Subway during the comparative audit. China Bar could use simple, image-based signs with clear examples of accepted and non-accepted items for each stream (e.g., general waste, recyclables), placed at eye level and directly on/above bins. Consistent signage across FOH, kitchen, prep, and dishwashing areas would improve staff accuracy, reduce contamination, and support cleaner, more efficient waste separation overall.



(Subway)



To start with, China Bar should install simple, picture-based bin signage on every waste station (like the example shown) so staff can quickly sort items correctly and reduce contamination.

A key limitation at China Bar is that the current collector (Urban Waste) does not provide pickup weight data for each service. This creates a measurement gap: without kg-per-pickup data, it is difficult to quantify waste generation, track trends, set reduction targets, or verify whether interventions are actually working.

By comparison, the JJ's Waste & Recycling invoice (used by Subway/Red Cup Café) shows weight-based records through "Excess Weight" entries for a 1100L wheeled bin (e.g., 66 kg, 46 kg, and 44 kg on different collection dates). Because these weights are recorded, those sites can monitor waste performance more accurately and identify opportunities to reduce disposal costs and contamination. This highlights a clear improvement opportunity for China Bar: request weight reporting from the contractor (or equivalent monthly data summaries) to support evidence-based waste management decisions.

PO BOX 235
CLEVELAND QLD 4163



A.B.N. 40 000 805 425

RED CUP CAFE
LINDA LI
1124 WHITEHORSE ROAD
BOX HILL VIC 3128

Invoice and Accounts Enquiries

Phone: (03) 9794 5722

Email: ar.melbourne@jjswaste.com.au

Melbourne Operational Enquiries

Phone: (03) 9794 5722

Email: operations.melbourne@jjswaste.com.au

Tax Invoice / Statement

Month: DEC-2025 Customer No: 09081440
Invoice No: 090814402512
 Issue Date: 31/12/2025

Total Due: \$889.50

PAY NOW

Account Summary (including GST)

Opening Balance: \$561.78
 Payments made this period: \$0.00_{cr}

Balance Overdue **\$561.78** - **DUE NOW**
Transactions for this period **\$327.72** - **Due By: 14/01/2026**

(Total GST for this invoice \$29.78)

Total Balance: **\$889.50**

METHODS TO PAY

TERMS STRICTLY 14 DAYS

**Access your
JJ's Customer Portal**

Go to:
myaccount.jjswaste.com.au
 and use your
 Customer Number to:

- Pay your Invoice
- View your account information
- Book / Cancel services
- Contact us



Billir Code: 28357
Ref: 090814401

Telephone & Internet Banking – BPAY®
 Contact your bank or financial institution to make this payment from your cheque, savings, debit, credit card or transaction account. More info: www.ibpay.com.au

Any payment must be for the exact amount of this invoice and be made by the due date. Otherwise, any amount paid will not be accepted and will be returned.

EFT

Acc Name: J.J. Richards & Sons Pty Ltd
 BSB: 064 168
 Acc #: 0045 0396
 Email remittance: ar.remit@jjswaste.com.au

Ref No: 09081440



PayID: 40000805425
 Email remittance: ar.remit@jjswaste.com.au

Ref No: 09081440

PayID is a registered trademark of NPP Australia Limited.

DIRECT DEBIT

Payments can be arranged for monthly automatic deductions on the 15th of each month from your nominated savings, cheque or credit card account.

Please phone our office to set-up Direct Debit as your payment option.

Transaction Details

SUPPLY OF CONTAINERS, REMOVAL & DISPOSAL OF
TRADE WASTE FOR THE PERIOD ENDING: 31/12/2025



A.B.N. 40 000 805 425

DATE	DESCRIPTION	DOCKET	PRICE	GST	TOTAL	BALANCE
	BALANCE BROUGHT FORWARD					561.78
OVERDUE BALANCE FOR PAYMENT						561.78

Overdue Balance

Your invoice shows an overdue balance, this needs to be paid immediately to avoid possible suspension of your services. Should your services be suspended you could incur an Overdue Fee. If you have a query regarding this balance please call our Accounts Department on (03) 9794 5722 to discuss your account.

Transactions for this period

1124 WHITEHORSE ROAD BOX HILL						
EXCESS WEIGHT CHARGE						
01/12/25	1 NA		25.06	2.51	27.57	
	EXCESS WEIGHT 66KG - 1100L WHEELED BIN					
08/12/25	1 SERVICE		16.52	1.65	18.17	
	EXCESS WEIGHT 46kg - 1100L WHEELED BIN					
22/12/25	1 SERVICE		15.66	1.57	17.23	
	EXCESS WEIGHT 44kg - 1100L WHEELED BIN					
1100L WHEELED BIN						
01/12/25	1 BIN		48.14	4.81	52.95	
08/12/25	1 BIN		48.14	4.81	52.95	
15/12/25	1 BIN		48.14	4.81	52.95	
22/12/25	1 BIN		48.14	4.81	52.95	
29/12/25	1 BIN		48.14	4.81	52.95	
	SUB-TOTAL		297.94	29.78	327.72	
Transactions for this period						327.72

Errors and Omissions Excepted

INVOICE NUMBER: **090814402512**

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At China Bar, oil residues are being collected separately. However, the container should be managed more safely: keep it clearly labelled, covered when not in use, placed in a stable spill-controlled area (away from stormwater/rain), and emptied on a regular schedule through an authorised oil collector. This turns a good practice into a strong, low-risk sustainability control.



(Red cup café)



(China Bar)

China Bar can implement a dedicated used cooking oil collection system similar to Red Cup Cafe by partnering with a licensed collector (e.g., a service like Your Oil Man), and using a clearly labelled, sealed “waste cooking oil only” drum in the back-of-house area. Staff should transfer cooled used oil from fryers/woks into this drum using a funnel/closed container.

Operationally, China Bar should assign one staff role per shift to monitor drum level, keep the lid closed, and arrange pickups before overflow. A simple pickup log (date, drum level, collector, and any spills/issues) can be used for tracking and reporting. They should also keep absorbent material/spill kit nearby and include a short SOP: “cool oil → transfer safely → keep lid closed → record level.”

At China Bar, keeping the 1100 L general waste bin and the waste-oil container outdoors (fully exposed to weather) can create several problems. Rainwater can enter bins and increase moisture, which accelerates decomposition, causes stronger odour, and attracts pests (flies/rodents). Wet waste can also produce leachate (dirty liquid) that may leak onto the ground and become a slip hazard for staff. In windy/rainy conditions, lids can be left open or moved, increasing litter escape and contamination risk around the back lane.

For the oil residue container, outdoor exposure is even riskier. If rain enters or the container is not tightly sealed, oil-water mixture can overflow or leak, creating slippery surfaces and possible stormwater pollution. This is both a hygiene/safety issue and an environmental compliance risk. Water-contaminated oil can also reduce recovery quality for recyclers and make collection less efficient.



(China Bar)

A better practice (as seen at Red Cup Café) is to keep these units in a covered, controlled waste storage area: roofed, impermeable floor, lids always closed, and easy but restricted access. If indoor placement is not possible, China Bar should install a weatherproof enclosed waste bay outside with: (1) lockable lids, (2) clear signage, (3) routine wash-down/inspection, and (4) secondary containment (bund/spill tray) for oil containers plus a nearby spill kit.

Bund/spill tray method:

It's a backup container/barrier that catches liquid if the main oil container leaks, tips, or overflows.

- Bund = a low walled area (like a mini “bathtub” around a drum/tank).
- Spill tray = a tray placed under the container to catch drips/leaks.

So if waste oil escapes, it stays in the bund/tray instead of going onto the floor or into drains.



(Red cup café)

Cardboard Management:

Red Cup Cafe:

Red Cup Cafe shows a good cardboard management practice by keeping flattened cardboard together in a designated indoor spot. This improves recycling quality, keeps cardboard dry, and prevents mixing with food/general waste. The setup is simple and practical for small businesses. A small improvement would be adding a clear “Cardboard Only” sign and tying stacks to keep the area tidy and safe.



Subway:

Subway also shows a good practice by using a separate storage space for cardboard, which supports source separation and keeps recyclables out of general waste. This lowers contamination risk and improves recycling outcomes. The main improvement would be flattening all boxes immediately (instead of storing bulky boxes) to save space and reduce handling time.



China Bar:

Since China Bar does not currently have a dedicated cardboard storage area, there is a higher risk of cardboard being mixed with general waste, getting contaminated, or becoming wet. This reduces recyclability and can increase collection costs and clutter. A practical solution is to copy the Red Cup/Subway model: create a clearly marked cardboard-only storage point near the prep/back area, flatten boxes at source, and transfer them on a set schedule to the cardboard stream (or cardboard pickup). If space is limited, use a vertical cage or slim rack to keep flattened cardboard dry, tidy, and separate.



Using paper bags instead of plastic bags is a positive sustainability step because paper is generally easier to recycle and causes less long-term litter impact.

To strengthen it further: make paper bags the default, and provide stronger “reusable plastic” bags only when needed (e.g., heavy, wet, or oily orders).



(paper bags)



(plastic bags)

At China Bar, having no lids on bins can lead to stronger odour, easier pest access (flies/rodents), and higher contamination risk because items can be dropped into the wrong bin more easily. For any bins near food-prep or dishwashing areas, open bins also increase hygiene risk from splash/aerosols and make the area look less controlled. For outdoor bins, no lid also means rainwater can enter, creating leachate, increasing smell, and making waste harder to handle.

A good practice would be to make “lidded bins by default” for all general waste and food-contact areas. Use foot-pedal lids indoors (hands-free = better hygiene), and ensure external bins have functional hinged lids that are kept shut after each use. Add a simple sticker near each bin: “Keep lid closed” and include this in opening/closing checklists. Assign one staff member per shift to do a quick bin check (lid closed, contamination, fill level, liner condition).

Practical exception: for clean cardboard-only storage, a lid is less critical if cardboard is kept dry, stacked neatly, and away from food areas. But for general waste and food-related bins, lids should be treated as essential.

Making compost out of waste ?



(coffee grounds)

At Red Cup Café, coffee grounds are easy to separate and low-risk, so giving them away for compost works well. For China Bar, food organics are more mixed (rice, sauces, meat, bones), so informal “take it home” collection is harder because of smell, leakage, pests, contamination, and staff transport issues. A better approach is to introduce a separate, lidded food-organics bin in the prep area and partner with a licensed organics collector/composter. Start with back-of-house prep scraps first (vegetable trims, food prep offcuts), then expand if contamination stays low.

I suggest “sell it (organic waste) and avoid landfill” , this idea is strong, but in practice, food organics usually reduce costs more than generate direct income. The main financial benefit is often lower general-waste volume and fewer landfill pickups, while the environmental benefit is major (less food waste to landfill).

If China Bar wants a community model, it should be limited to clean streams only (for example, vegetable offcuts), with sealed tubs and clear pickup rules. But this model is hard to run in real life.

At China Bar, FOH staff currently enter orders on an iPad, but each order is printed in the kitchen as a paper KOT. This creates significant paper waste every day, especially during busy service periods.

To reduce waste and improve efficiency, I recommend extending the digital ordering system to the kitchen by installing kitchen display iPads. Once FOH enters an order, it can appear instantly on the relevant kitchen screens instead of being printed. This would cut daily paper use, reduce printing costs, and lower the risk of lost or unclear tickets.

I also recommend switching to e-receipts as the default option, with paper receipts provided only on request. This approach maintains customer convenience while further reducing unnecessary paper consumption.

For smooth kitchen operations, separate display screens should be assigned by station/course (e.g., drinks, mains, BBQ). This keeps workflow organised, prevents congestion on a single screen, and supports faster, more accurate order preparation.

STAFF TRAINING

Method: “10–10–10 Waste Training System”

1) 10 minutes induction (for every new staff member)

Do this on their first shift (or before they start):

- Show the bin map (FOH, dishwash, kitchen, outside bins) + what each bin is for.
- Explain the “Top 10 wrong items” you’ve observed (e.g., milk cartons in general waste, food in cardboard, bottles mixed incorrectly).
- 1-minute rule: “If you’re unsure, ask the shift leader—don’t guess.”
- Finish with a 60-second quiz: “Where does X go?” (5 quick items).

2) 10-minute toolbox talk weekly (everyone on shift)

Once a week, just before service (or after close):

- Pick one problem only (example: “stop bathroom tissue going to dishwasher bin”).
- Explain why it matters (contamination, hygiene, cost).
- Show the correct action physically (walk to bin + demonstrate).
- Give one clear rule (e.g., “bathroom bins always get a fresh liner; ties closed; never emptied into BOH bins”).

3) 10-second reminders at the bin (“training at point of use”)

This is the most powerful part:

- Put simple signage on each bin with pictures/examples.
- Add 3 lines max:
 - YES: (items allowed)
 - NO: (common wrong items)
 - IF UNSURE: ask supervisor

Assign a “Waste Champion” per shift

One person each shift (FOH + kitchen) checks:

- bins are not overflowing
 - liners and lids are used properly
 - contamination is corrected early
- This takes 2 minutes per check and stops problems before they spread.

Assign a free shared meal for the waste champions of the week, when weekly waste checks are completed and contamination targets are met.

Common hardships and how to overcome them

Staff resistance / “too busy” mindset

Challenge: During rush periods, staff may ignore sorting rules.

Fix: Use ultra-short training (10 minutes), bin stickers at point-of-use, and one simple rule per week. Make the Waste Champion role shift-based, not permanent.

High staff turnover

Challenge: New staff won’t know the system.

Fix: Add a 1-page waste SOP to onboarding + 5-item “where does this go?” induction quiz on day one.

Space constraints (especially BOH/prep areas)

Challenge: Hard to add more bins/storage zones.

Fix: Use slim bins, stackable crates, vertical cardboard racks, and redesign layout so each station has the minimum required streams.

Contamination from unclear sorting

Challenge: Wrong items in bins reduce recovery and increase landfill.

Fix: Standardised signage (photos of real site items), color coding, and weekly 5-minute contamination spot checks.

No reliable weight data from contractor

Challenge: Hard to measure progress and prove outcomes.

Fix: Track proxy metrics: bin fullness scores, contamination incidents, number of CDS containers returned, cardboard stack volume, pickup frequency, overflow events.

Upfront costs (lids, signage, extra bins, training time)

Challenge: Management may delay actions due to cost.

Fix: Phase rollout: start with low-cost quick wins first (signage, SOPs, lid policy, cardboard zone), then scale to higher-cost items if results improve.

Outdoor exposure (rain, wind, leaks, odour)

Challenge: Bins/oil containers outside can get wet and create hazards.

Fix: Covered waste bay, closed lids, spill trays/bunds for oil, and routine inspection checklist.

Customer behavior (wrong disposal, leftovers, bones on tables)

Challenge: FOH waste quality becomes inconsistent.

Fix: Clear customer-facing prompts, faster table clearing, and FOH staff correction at source.

Inconsistent follow-through after first month

Challenge: Good systems fade without accountability.

Fix: Monthly review meeting (10–15 min), assign KPI (Key Performance Indicator) owner, publish a simple dashboard on staff noticeboard.

Unclear contractor service scope

Challenge: Missed opportunities for glass/cardboard/organics optimization.

Fix: Quarterly service review with contractor; request documented service matrix (what accepted, pickup frequency, contamination policy, optional streams).

CONCLUSION

This waste assessment was conducted to move China Bar's waste management from assumption-based decisions to evidence-based action. The purpose was not only to identify what is being discarded, but to understand *why* waste is occurring at point-of-generation (FOH, prep, dishwashing, kitchen), where contamination enters the system, and which operational practices are helping or limiting diversion. Although the assignment required one organisation, comparing practices across China Bar, Subway, and Red Cup Café provided a stronger benchmark for practical improvements that are realistic in food-service settings.

The data shows that China Bar already has several positive foundations (cardboard separation behaviour by staff, CDS-related bottle collection, reusable dine-in cutlery, and willingness to engage with contractor information). However, the audit also identified clear structural gaps: no bin signage, no lids, inconsistent stream separation at source, informal cardboard storage, outdoor exposure risks for major bins/oil residue, and limited contractor reporting (no pickup weight data). In practical terms, this means the current system depends heavily on staff memory and effort rather than a controlled process. As a result, contamination risk remains high, recoverable materials are lost to general waste, hygiene and safety risks increase, and performance is difficult to measure over time.

Therefore, the most effective strategy is to redesign the system around simple controls: standardised signage on every bin, lidded bins for all food-contact/general streams, dedicated and dry cardboard storage, clear CDS vs non-CDS separation, formal oil handling controls, and structured staff training with shift-level accountability. Together, these actions will improve source separation quality, reduce landfill-bound waste, improve hygiene/safety outcomes, and create a more auditable waste system. The long-term success of this plan depends on one final enabler: routine data tracking (service records, contamination checks, fill-level trends, and contractor reporting), so improvements can be verified rather than assumed.

Timeline to achieve recommendations

- **0–30 days (Immediate controls):**
Install bin signage site-wide; implement “keep lid closed” rule with lidded bins where required; start FOH/BOH waste briefing; set up clear CDS/non-CDS container separation; assign one Waste Champion per shift.
- **31–90 days (System stabilisation):**
Formalise cardboard-only storage zone; introduce restroom and dishwashing waste handling SOPs; improve outdoor waste/oil area controls (weather protection + spill containment); run weekly 5-minute contamination checks; begin monthly contractor review meetings.
- **3–6 months (Performance phase):**
Negotiate improved contractor reporting (weights/volume proxies where possible); trial organics separation for clean prep scraps; review pickup frequencies using observed fill patterns; evaluate cost, contamination trend, and diversion progress against baseline.
- **6+ months (Continuous improvement):**
Lock in successful practices as standard operating procedure, refresh staff training quarterly, and update targets annually to drive ongoing reduction in landfill waste and contamination.

